

GROUP PROJECT

QuizCraft

Smart Quiz Generator

Automatically creates MCQ quizzes from study notes or PDF content.

Team: Tamanna • Vanshika • Team Member

My Role: Development, testing, UI and documentation

Presentation • 5–8 min

Problem Statement

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The Problem

Creating MCQ quizzes manually from notes and PDF study material takes time. Questions also need to be varied, relevant and easy to organize.

Why It Matters

Students need quick practice and teachers need a faster way to prepare assessment material. QuizCraft reduces repetitive manual work.

One-Sentence Solution

QuizCraft takes text or a PDF, processes the study material, extracts useful concepts and generates a quiz with multiple-choice questions.

Simple Flow

1. Enter topic / study material
2. Process text
3. Generate MCQs
4. Attempt quiz
5. Submit and view score
6. Save quiz history

Technology Stack

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Python

Core programming language

Streamlit

Web interface and app flow

SQLite

Users and quiz history

PyPDF2

Extracts text from PDFs

Pandas

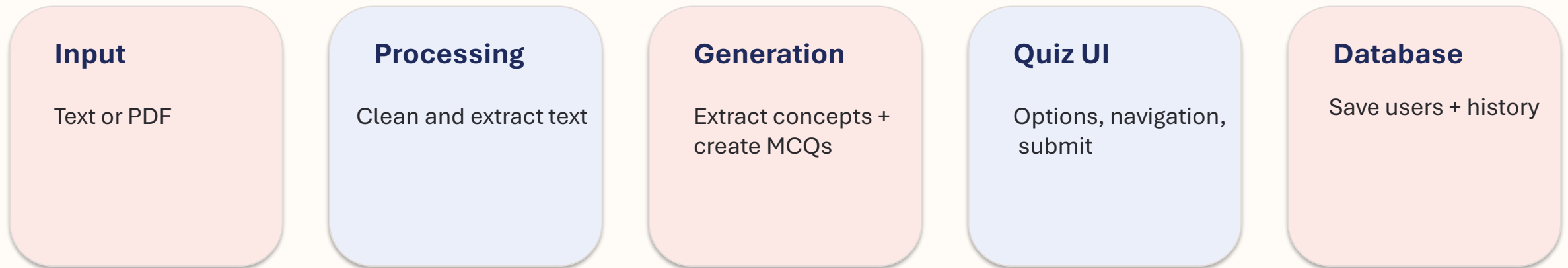
Data handling

Regex Pattern

Indetify text patterns

Architecture — How It Works

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User → Input → Text Processing → Quiz Generation → Streamlit Quiz → Score → SQLite History

LIVE DEMO — Show the App Running

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QUIZCRAFT — LIVE APPLICATION

Login / Sign Up

Create Quiz

Topic + Difficulty

MCQ + 4 Options

Next / Submit

Score + Performance + History

<https://quizcraft-ht5u2jowmcr32d594db23a.streamlit.app/>

Click the link to view the live demo

Current Approach

No external AI API is required in the current version. Python-based text processing and rule/pattern-based concept extraction are used to create questions.

Simple Logic

1. Extract text from notes/PDF
2. Clean unnecessary spaces
3. Find concept patterns such as definitions and uses
4. Build question + options
5. Display the quiz in Streamlit

Challenges Faced

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Challenge 1:

Designing a user-friendly interface in Streamlit for QuizCraft.

Challenge 2:

Managing page navigation and interactive elements smoothly in QuizCraft

My Role — What I Did

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My contribution to the group project:

Front-End Development

Worked on the Streamlit front-end development of QuizCraft – Smart Quiz Generator.

Designed the user interface, page layout, and interactive elements to make QuizCraft simple and user-friendly.

Future Improvements

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- Use stronger NLP or LLM-based question generation.
- Add True/False and fill-in-the-blanks questions.
- Improve distractor quality for more challenging options.
- Add topic-wise and difficulty-wise performance analytics.
- Support more document formats and multilingual quizzes.

GitHub Repository

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Project Repository

GitHub contains the project source code, requirements, README, sprint documentation and supporting files.

<https://github.com/vanshikarathore910-max/Quizcraft.py.git>

Included Files

app.py
quiz_generator.py
text_processor.py
database.py
requirements.txt
README.md / SPRINTS.md

THANK YOU

QuizCraft — Smart Quiz Generator

Questions & Answers

Thank you for listening to our presentation.